

1. Let A be an $n \times m$ matrix.

- (a) (3 points) Show that $\ker(A^T A) = \ker(A)$. Hint: show that every vector in the left kernel is also in the right kernel, then vice versa.
- (b) (2 points) Show that when the columns vectors of A are linearly independent, $A^T A$ is invertible.

(a) proof ①:

Let $\vec{x} \in \ker(A)$, $A\vec{x} = \vec{0}$. Then,

$$\begin{aligned} A^T A \vec{x} &= A^T (A \vec{x}) \\ &= A^T (\vec{0}) \\ &= \vec{0} \end{aligned}$$

so $\vec{x} \in \ker(A^T A)$

Since any $\vec{x}$ in $\ker(A)$ is also in $\ker(A^T A)$,

$$\boxed{\ker(A) \subseteq \ker(A^T A)}$$

proof ②: Let $\vec{x} \in \ker(A^T A \vec{x})$, so

$$A^T A \vec{x} = \vec{0}$$

$$\Rightarrow \vec{x}^T (A^T A \vec{x}) = \vec{x}^T (\vec{0})$$

$$\begin{aligned} \vec{x}^T (A^T A \vec{x}) &= (\vec{x}^T A^T) (A \vec{x}) \\ &= (\vec{x} A)^T (A \vec{x}) \end{aligned}$$

$$= \|A \vec{x}\|^2 = 0$$

This means $A \vec{x} = \vec{0}$

$\vec{x} \in \ker(A)$

so $x \in \ker(A^T)$

So any $\vec{x}$ in $\ker(A^T A)$ is also in $\ker(A)$,

$$\boxed{\ker(A^T A) \subseteq \ker(A)}$$

Therefore, combining proof ① & ②, $\ker(A^T A) = \ker(A)$

(b) Since column vectors of A are LI, then

$$\ker(A) = \{\vec{0}\}$$

From part (a), $\ker(A^T A) = \ker(A)$ &
 $\ker(A) = \{\vec{0}\}$, so $\ker(A^T A) = \{\vec{0}\}$.

$A^T A$ is an $m \times m$ matrix w/ trivial kernel
($m \times n$) ($n \times m$)

or LI cols.

A square matrix Q is invertible iff $\ker(Q) = \{\vec{0}\}$.

Since $A^T A$ is $m \times m$, and $\ker(A^T A) = \{\vec{0}\}$,

we can conclude that $A^T A$ is invertible.

2. (4 points) Let $P = \begin{bmatrix} 1 & 1 & 1 & 1 \\ k & -2 & -1 & 2 \\ 1 & 4 & 1 & 4 \\ -3k & 0 & 3 & -6 \end{bmatrix}$. Calculate the determinant of P . Hint: your answer will involve the constant k .

$$\det(P) = a_{11}|A_{11}| - a_{12}|A_{12}| + a_{13}|A_{13}| - \dots + (-1)^{1+n}a_{1n}|A_{1n}|$$

$$C_{ij} = (-1)^{i+j}|A_{ij}|$$

$$\det(P) = (1) \cdot C_{11} - (1) C_{12} + (1) C_{13} - (1) \cdot C_{14}$$

$$C_{11} = (-1)^{1+1} |A_{11}|$$

$$|A_{11}| = 1 \cdot \begin{vmatrix} -2 & -1 & 2 \\ 4 & 1 & 4 \\ 0 & 3 & -6 \end{vmatrix} \quad \leftarrow \text{expand along row 3}$$

$$= (0) \cdot \begin{vmatrix} -1 & 2 \\ 1 & 4 \end{vmatrix} - (3) \begin{vmatrix} -2 & 2 \\ 4 & 4 \end{vmatrix} + (-6) \cdot \begin{vmatrix} -2 & -1 \\ 4 & 1 \end{vmatrix}$$

$$= 0 - 3(-8-8) + (-6)(-2+4)$$

$$= 0 + 48 - 12 = \boxed{36} \leftarrow C_{11}$$

$$C_{12} = (-1)^{1+2} |A_{12}| = (-1) |A_{12}|$$

$$|A_{12}| = \begin{vmatrix} k & -1 & 2 \\ 1 & 1 & 4 \\ -3k & 3 & -6 \end{vmatrix} \quad \leftarrow \text{expand along row 1}$$

$$= k \cdot \begin{vmatrix} 1 & 4 \\ 3 & -6 \end{vmatrix} - (-1) \cdot \begin{vmatrix} 1 & 4 \\ -3k & -6 \end{vmatrix} + 2 \cdot \begin{vmatrix} 1 & 1 \\ -3k & 3 \end{vmatrix}$$

$$= k(-6+12) + (-6-12k) + 2(3+3k)$$

$$= -18k + 12k - 6 + 6 + 6k = 0$$

$$C_{12} = (-1) \cdot 0 = 0$$

$$C_{13} = (-1)^{1+3} |A_{13}| = (1) |A_{13}|$$

$$|A_{13}| = \begin{vmatrix} k & -2 & 2 \\ 1 & 4 & 4 \\ -3k & 0 & -6 \end{vmatrix} \leftarrow \text{expand along row 3}$$

$$= (-3k) \cdot \begin{vmatrix} -2 & 2 \\ 4 & 4 \end{vmatrix} - (0) \cdot \begin{vmatrix} k & 2 \\ 1 & 4 \end{vmatrix} + (-6) \begin{vmatrix} k & -2 \\ 1 & 4 \end{vmatrix}$$

$$= (-3k)(-8-8) - 0 + (-6)(4k+2)$$

$$= 48k - 24k - 12$$

$$= 24k - 12$$

$$C_{13} = 24k - 12$$

$$C_{14} = (-1)^{1+4} |A_{14}| = (-1) |A_{14}|$$

$$|A_{14}| = \begin{vmatrix} k & -2 & -1 \\ 1 & 4 & 1 \\ -3k & 0 & 3 \end{vmatrix} \leftarrow \text{expand along row 3}$$

$$= (-3k) \begin{vmatrix} -2 & -1 \\ 4 & 1 \end{vmatrix} - (0) \begin{vmatrix} k & -1 \\ 1 & 1 \end{vmatrix} + (3) \begin{vmatrix} k & -2 \\ 1 & 4 \end{vmatrix}$$

$$= (-3k)(-2+4) - 0 + 3(4k+2)$$

$$= -6k + 12k + 6 = 6k + 6$$

$$C_{14} = -6k - 6$$

$$\det(p) = C_{11} + C_{12} + C_{13} + C_{14}$$

$$= 1 + 0 + 24k - 12 + (-6k - 6)$$

$$= 36 + 0 + 24k - 6k - 6 = 18 + 18k.$$

3. (2 points each)

(a) Suppose $\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = 2$. Evaluate the following determinant:

$$\begin{vmatrix} 2d & 2e & 2f \\ a & b & c \\ -5g & -5h & -5i \end{vmatrix}.$$

(b) Suppose A is a 5×5 matrix with $|A| = 3$, and let $k \in \mathbb{R}$ be a real number. What is the determinant of the matrix $k(A^{-2})^T A$? *Hint: your answer will involve the constant k .*

(a)